



SAAD ZIAATHAR KHAN

B.E. - Computer Engineering

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PROFESSIONAL SUMMARY

Innovative Full-Stack Developer with 2+ years of hands-on experience in building scalable web solutions, AI/ML pipelines, and cloud-native architectures. Skilled in JavaScript, TypeScript, Python, React, Next.js, and LangChain, with a strong track record of delivering modern applications that combine performance, usability, and intelligent automation. Experienced in designing responsive frontend interfaces, developing robust backend services, and integrating APIs, databases, and third-party platforms to create seamless end-to-end digital products. Adept at translating business requirements into practical technical solutions and shipping features that improve user engagement, system reliability, and operational efficiency.

Throughout my development experience, I have worked on a variety of projects involving full-stack web development, real-time systems, data-driven applications, and AI-enhanced workflows. I have built user-centric interfaces with React and Next.js, implemented secure and scalable backend logic using Python and Node.js, and developed retrieval and automation features using LangChain and modern LLM-based tools. My work also includes integrating cloud services, optimizing application performance, and ensuring maintainable code through modular architecture and clean development practices.

I am comfortable working across the complete product lifecycle, from planning and prototyping to implementation, testing, deployment, and iteration. I have collaborated with cross-functional teams to deliver feature-rich applications under tight timelines, while maintaining attention to code quality and user experience. My approach combines strong problem-solving ability, continuous learning, and a practical understanding of emerging technologies to build solutions that are both innovative and production-ready.

Driven by a passion for creating impactful digital products, I aim to develop applications that solve real-world problems through smart engineering and thoughtful design. My experience in web development, AI integration, and cloud-native systems has helped me build a solid foundation for creating scalable, efficient, and future-ready software solutions.

KEY EXPERTISE

C++ Java Python Spring Boot C HTML CSS Javascript ReactJS NodeJS SQL MongoDB
Machine Learning Data Structures Algorithms Git Github Linux REST TailwindCSS Bootstrap
Problem Solving Object-Oriented Programming DBMS Operating Systems Computer Vision Data Analysis NumPy
Pandas Scikit-Learn Matplotlib Web Development React Native Expo Langchain HuggingFace OpenAI
Gemini

EDUCATION

Sinhgad College of Engineering

2023 - 2027

B.E. - Computer Engineering | CGPA: 8.07 / 10

School of Scholars, Nagpur

2023

12th | CBSE | Percentage: 91.40 / 100

D. A. V. Public School, Wani

2021

10th | CBSE | Percentage: 94.40 / 100

PROFESSIONAL EXPERIENCE

InvizioX | IT / Computers - Software

25 Jan, 2026 - 25 Jun, 2026

Mobile Application Developer

Key Skills: React Native expo EAS Testing NodeJS NativewindCSS CSS Javascript Android Studio

Engineered and developed InvizioX, an AI-powered asset management platform supporting 3+ asset classes, including stocks, cryptocurrency, and commodities, with intelligent shell-based portfolio organization and automated balancing logic. Built responsive React Native interfaces and analytics dashboards that improved portfolio monitoring efficiency by 40%, while integrating ML/DL-driven predictive models and blockchain-backed verification to enhance risk analysis, transaction transparency, and secure asset reallocation workflows. Successfully designed and delivered 10+ core modules, including shell management, balancing engine, trend prediction, and real-time portfolio visualization.

EntrepreneurEdge | IT / Computers - Software

26 Jul, 2024 - 15 Jan, 2025

Frontend Web Developer

Key Skills: HTML5 CSS Javascript NodeJS Express.js ReactJS TailwindCSS NextJS

Engineered highly responsive React interfaces that enhanced user engagement by 35% through intuitive and user-centric design. Optimized the codebase and improved application performance, reducing page load times by 40% across critical platforms. Successfully accelerated development cycles by delivering an MVP within just three weeks while consistently exceeding client expectations and project requirements.

E-CELL SCOE | IT / Computers - Software 10 Nov, 2023 - 12 Jun, 2024

Full Stack Web Developer

Key Skills:

HTML CSS GSAP Flask NodeJS ReactJS React Native Next.js Express.js Javascript Python
Docker SQL MongoDB Firebase

Architected and developed web platforms for more than three startups, driving a 45% increase in user engagement and a 30% boost in conversion rates. Led automation-focused initiatives that improved operational efficiency and expanded outreach by 30%, while also facilitating cross-functional collaboration across teams to successfully deliver four product launches and over ten feature enhancements.

INTERNSHIPS

InvizioX | IT / Computers - Software 25 Jan, 2026 - 25 Jun, 2026

Mobile Application Developer

Key Skills: React Native CSS Javascript NativewindCSS Expo Github Git API EAS

Engineered and developed Inviziox, an AI-powered asset management platform supporting 3+ asset classes, including stocks, cryptocurrency, and commodities, with intelligent shell-based portfolio organization and automated balancing logic. Built responsive React Native interfaces and analytics dashboards that improved portfolio monitoring efficiency by 40%, while integrating ML/DL-driven predictive models and blockchain-backed verification to enhance risk analysis, transaction transparency, and secure asset reallocation workflows. Successfully designed and delivered 10+ core modules, including shell management, balancing engine, trend prediction, and real-time portfolio visualization.

EntrepreneurEdge | IT / Computers - Software 26 Jul, 2024 - 15 Jun, 2025

Frontend Web Developer

Key Skills:

Git Javascript SQL Three.js HTML5 NextJS GSAP Express.js Firebase TailwindCSS NodeJS CSS
Github ReactJS

Engineered highly responsive React interfaces that enhanced user engagement by 35% through intuitive and user-centric design. Optimized the codebase and improved application performance, reducing page load times by 40% across critical platforms. Successfully accelerated development cycles by delivering an MVP within just three weeks while consistently exceeding client expectations and project requirements.

PROJECTS

Emergency Alert System for Senior Citizens

Key Skills:

HTML CSS Javascript ReactJS NodeJS Express.js NextJS TailwindCSS Python API Sockets
Servers Geolocation API Open Street Map Open Route Services

Project Link: <https://github.com/SaadZiaatharKhan/CareConnect-search-nurses-near-you>

CareConnect is a real-time healthcare assistance and emergency response platform designed to support senior citizens living independently by connecting them with nearby nurses during medical emergencies. The system combines AI-powered healthcare support, geolocation tracking, and automated emergency escalation to improve response efficiency and patient safety.

The platform tracks the live location of 50+ active nurses using Geolocation APIs and identifies the nearest available responder through geo-distance calculations using the Haversine Formula and Open Route Service APIs. In emergency situations, the system instantly sends alerts to the closest nurse through WebSocket-based real-time communication. If no response is received within a predefined timeframe, the request is automatically escalated to the next nearest nurse, improving emergency response reliability by an estimated 60%.

CareConnect also integrates an AI-powered chatbot using the Gemini API, capable of processing both text and image inputs for healthcare-related assistance. Users can ask medical questions, upload symptom-related images, and receive intelligent support instantly. Additionally, the platform includes a voice-enabled reminder alarm system that helps patients manage medication schedules, doctor appointments, and daily healthcare routines using Speech APIs.

The application was developed using Next.js, React, TailwindCSS, Node.js, WebSockets, OpenStreetMap, and Open Route Service APIs. Server-side rendering (SSR) and dynamic imports were implemented to improve SEO optimization and reduce application load time by nearly 40%. The system features a responsive and user-friendly interface optimized for both desktop and mobile devices.

Overall, CareConnect demonstrates strong expertise in full-stack development, real-time systems, AI integration, geolocation services, and healthcare-focused software engineering. The project successfully combines emergency response automation, intelligent healthcare assistance, and scalable modern web technologies into a socially impactful digital healthcare solution.

AgriGuardian: AI Crop Disease Detection & Treatment Advisor

Key Skills:

React Native NativewindCSS CSS Expo NodeJS Python FastAPI Computer Vision OpenCV Langchain
Gemini HuggingFace Supabase SerpAPI Scikit-Learn Arduino ESP Embedded C Hardware
Machine Learning

Project Link: <https://github.com/SaadZiaatharKhan/AgriGuardian>

AgriGuardian is an AI-powered crop disease detection and treatment advisory platform designed to help farmers identify plant diseases at an early stage, reduce crop loss, and improve farming efficiency through real-time decision support. The system combines 3 advanced CNN models, VGG-16, Inception V3, and ResNet-50, with OpenCV-based image preprocessing to analyze crop images captured through an ESP32-CAM module. With this pipeline, the project achieves up to 94% accuracy in crop disease detection and supports fast, image-based diagnosis in field conditions. It also processes environmental inputs from 6+ IoT sensors, including soil moisture, rain, flame, LDR, DHT22 temperature and humidity, and other hardware signals, so that disease prediction is not based only on images but also on crop surroundings. The weather forecasting module integrates Open-Meteo data and supports up to 99% accuracy in weather prediction, helping farmers make better irrigation and disease-prevention decisions.

The platform was developed as a cross-platform mobile application using React Native, Expo, NativeWindCSS, Supabase, Node.js, FastAPI, and Python, with TensorFlow, Scikit-learn, LangChain, Gemini, and OpenCV used for intelligence and automation. It provides multilingual support in 15 native languages, making the system accessible to farmers in remote and diverse regions. The application also includes automated image capture, real-time disease alerts, tailored treatment suggestions, and nearby agro-shop recommendations through Google Maps API. In addition, the system can suggest optimal selling prices and support remote monitoring, which makes it useful for both small and large-scale farming operations. Overall, AgriGuardian is a practical smart agriculture solution that combines AI, IoT, and generative intelligence to reduce pesticide usage, improve crop health, and support sustainable farming with measurable, technology-driven impact.

Document Intelligence Hub

Key Skills:

HTML CSS Javascript ReactJS NodeJS Express.js NextJS TailwindCSS TypeScript Bootstrap
Supabase Machine Learning Python FastAPI MySQL FAISS Vector Database Langchain HuggingFace
OCR Open Source Software RAG

Project Link: <https://github.com/SaadZiaatharKhan/KMRL>

Document Intelligence Hub is a project built around an offline multimodal Retrieval-Augmented Generation system. Its main goal is to ingest, index, and query 4 different data formats, DOC, PDF, images, and voice recordings, inside one unified semantic retrieval framework. Instead of using separate tools for each file type, the system brings all sources into a shared vector space so users can search across text, images, and audio from a single interface. This makes the solution especially useful in sensitive or restricted

environments where secure offline access and citation-backed answers are important.

The technical pipeline starts with file upload and preprocessing, where text extraction, data segmentation, chunking, vectorization, OCR, and speech-to-text are used to convert heterogeneous content into searchable embeddings. The backend is built with Python, FastAPI, and Ollama, while the frontend uses React, TailwindCSS, and Electron. The storage layer combines MySQL with FAISS as the vector database, and the system also uses LangChain, Hugging Face, and Whisper for retrieval and multimodal processing. For generation, the project relies on 2 open-source LLMs, Mistral and Gemma, which support offline operation and reduce dependency on cloud services.

The project emphasizes practical impact through faster access, time savings, grounded outputs, and scalable design. It reduces manual searching across documents, screenshots, and audio segments by returning relevant information with exact citations, improving transparency and trust. The feasibility study highlights that mature core technologies such as OCR, embeddings, vector databases, and offline LLMs are already available, while the main challenges are data quality, storage needs, model hallucinations, and cross-modal alignment. Even with these constraints, the modular architecture supports future expansion to new data types, models, and analytics, making Document Intelligence Hub a strong prototype for secure, cost-effective, offline knowledge retrieval.

EchoGuide: AI-Powered Navigation for the Visually Impaired

Key Skills:

Python FastAPI YOLO Computer Vision Machine Learning Object Detection Models Real-time Object Detection Ambient Noise Detection ESP Hardware

Project Link: <https://github.com/SaadZiaatharkhan/EchoGuide>

EchoGuide is an AI-powered wearable navigation solution designed for visually impaired users who need real-time awareness of their surroundings in everyday environments. The project uses an ESP32-CAM module to capture the scene in front of the user and sends the data to a Python FastAPI server, where a lightweight object detection model such as MobileNet SSD or Tiny YOLO identifies nearby objects and estimates their distance. The system then converts this information into simple audio cues through a speaker, helping the user recognize obstacles like walls, chairs, doors, stairs, or even small items such as water bottles without depending on expensive assistive hardware. A microphone is also integrated for voice commands, allowing the user to ask location-based questions such as where a specific object is, while ambient noise monitoring helps adjust speaker volume automatically for clearer feedback. The project is especially relevant because more than 250 million people struggle with navigation, and many existing solutions remain costly or difficult to access.

The solution was designed to be affordable, portable, and scalable, making it suitable for wider adoption by individuals, caregivers, NGOs, schools, workplaces, and government programs. Its technical approach combines computer vision, machine learning, hardware integration, and server-based processing to keep the wearable device lightweight while offloading heavy computation to the backend. The system supports live object alerts with distance feedback such as “wall 1 m ahead” or “chair 0.7 m left,” and it can respond to voice queries using the latest camera frame. In testing, lightweight CV models can run in under 100 ms per frame on a Jetson Nano, while the ESP32 streams data reliably over Wi-Fi. A single-cell battery can power the device for more than 10 hours of continuous use, with the speaker and microphone consuming power only when active. This balance of speed, low cost, and portability makes EchoGuide a practical assistive technology with strong real-world impact, helping users move independently and confidently in unfamiliar spaces.

3D Zoo

Key Skills: ReactJS Javascript React Three Fiber React Three Drei Three.js Blender Web Hosting

Project Link: <https://saad-zoo.netlify.app/>

This Zoo is an interactive 3D web experience built using React Three Fiber and React Three Drei, designed to present a visually rich virtual animal environment inside a browser. The project uses Blender to combine and prepare the 3D models before loading them into the scene, allowing the zoo to feel more organized, realistic, and immersive. The website includes a total of 6 animated animals: Alpaca, Deer, Bull, Donkey, Fox, and Wolf. Each animal is displayed as part of a live 3D environment, making the project both visually engaging and technically strong in terms of model integration and animation handling.

The main focus of the project is to demonstrate how 3D assets can be rendered efficiently on the web using modern React-based graphics libraries. React Three Fiber provides the structure for building the 3D scene, while React Three Drei adds useful helper components that simplify camera controls, lighting, loading, and scene management. The models were first prepared and combined in Blender, which helped in reducing complexity and making the zoo environment easier to manage as a single 3D scene. This workflow shows a complete pipeline from model creation to browser deployment.

The project is also available in 2 versions: one without optimization and one with optimization. This makes it useful for comparing performance, loading speed, and rendering efficiency in real-world browser conditions. By maintaining 6 fully animated animals and multiple 3D assets, the zoo demonstrates how interactive web graphics can be built for educational, entertainment, or portfolio purposes. The website can be accessed directly through the deployed Netlify links, making it easy to view and test across different devices. Overall, the project highlights skills in 3D web development, model optimization, animation handling, scene composition, and modern frontend implementation, all within a single interactive zoo experience.

ACHIEVEMENTS

- o Solved 500+ questions of DSA across all platforms.
- o Participated in 7 National and International Hackathons.
- o Developed and deployed full-stack web applications using React, Next.js, Node.js, and Python, focusing on scalability, performance, and responsive user experience.

- o Architected AI-powered solutions integrating Machine Learning, Computer Vision, LangChain, and Generative AI technologies, achieving up to 94% prediction and detection accuracy across real-world projects.
- o Led cross-functional AI and full-stack projects using React, Next.js, Python, FastAPI, TensorFlow, LangChain, OpenCV, Supabase, and cloud-native technologies.

SEMINARS / TRAININGS / WORKSHOPS

Arduino Workshop Institute Name: Sinhgad College of Engineering

Key Skills: Arduino Hardware ESP

The Arduino workshop held at Sinhgad College of Engineering was a valuable hands-on learning experience designed to introduce students to embedded systems, electronics, and practical hardware programming. The workshop focused on helping participants understand the fundamentals of Arduino boards, sensors, actuators, and circuit connections through interactive demonstrations and guided exercises. It provided a strong foundation in microcontroller-based development and encouraged students to move beyond theory by building and testing real circuits.

During the workshop, participants learned how to interface Arduino with components such as LEDs, buzzers, motors, sensors, and display modules. They also gained exposure to the Arduino IDE, basic programming concepts, input-output operations, and the logic behind writing efficient code for hardware control. Through live examples and problem-solving activities, students were able to understand how Arduino is used in real-world applications such as automation, robotics, home control systems, and smart devices.

The workshop was highly engaging and skill-oriented, allowing students to work collaboratively while developing creativity, technical confidence, and practical problem-solving abilities. It also helped bridge the gap between classroom learning and industry-relevant application by showing how simple electronic components can be integrated into intelligent systems. Overall, the Arduino workshop at Sinhgad College of Engineering was an informative and inspiring event that strengthened the participants' understanding of electronics, coding, and embedded system design. It encouraged innovation and motivated students to explore further opportunities in the fields of IoT, robotics, and hardware development.

Web Development Workshop

Institute Name: BeTechnos

Key Skills: HTML CSS Javascript ReactJS

Completed a Web Development Workshop at BeTechnos Technologies focused on modern frontend and backend development technologies, gaining practical experience in designing responsive and interactive web applications. The workshop covered core web technologies including HTML, CSS, JavaScript, React.js, Node.js, and database integration, along with concepts related to responsive UI design, component-based architecture, and client-server communication.

During the workshop, worked on building dynamic web interfaces and implementing real-time functionalities using modern development practices. Developed hands-on experience with frontend frameworks, routing, API integration, form handling, and responsive layouts optimized for multiple screen sizes. The training also included project-based learning, debugging techniques, deployment workflows, and performance optimization methods used in production-ready applications.

The program emphasized practical implementation through mini-projects, collaborative development tasks, and live coding sessions, helping strengthen problem-solving and software engineering skills. Successfully completed multiple assignments and web modules while improving understanding of full-stack application architecture, user experience design, and scalable web development practices. The workshop enhanced technical proficiency in modern web technologies and provided industry-oriented exposure to professional development workflows and software project implementation.

CO-CURRICULAR ACTIVITIES

- o Participated in several national and international hackathons and coding competitions.
- o Developed AI-powered healthcare, agriculture, and accessibility projects.
- o Worked on real-time systems using React, FastAPI, TensorFlow, and LangChain.
- o Attended several workshops and technical sessions, including an international event organized by Red Hat.

EXTRA CURRICULAR ACTIVITIES

- o Volunteered with Prayas NGO to teach computer basics and modern technology to children in rural and underdeveloped communities, helping improve digital awareness and access.
- o Organized free medical camps to support community health initiatives and improve access to basic healthcare services for underserved groups.
- o Donated blood to contribute to emergency medical needs and support life-saving healthcare efforts.
- o Actively participated in college fest activities, contributing to event coordination, team collaboration, and smooth execution of programs.
- o Helped organize workshops and awareness campaigns to promote learning, engagement, and community outreach on important topics.

PERSONAL INTERESTS / HOBBIES

- Exploring new technologies and building personal projects to strengthen problem-solving and technical skills.
- Reading about AI, web development, and emerging software trends to stay updated with modern technology.
- Participating in coding challenges and hackathons to improve creativity, teamwork, and practical development experience.

WEB LINKS

- Personal - <https://saad-portfolio-olive.vercel.app/>
- Github - <https://github.com/SaadZiaatharKhan>

PERSONAL DETAILS

Gender: Male

Marital Status: Single

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